

Chris Wan . Jarvis Consulting

I am a graduate from the University of Toronto with a degree in Computer Science. My academic journey has equipped me with a strong foundation in programming and software principles such as clean code, maintainability, and effective debugging. Professionally, I have interned at Autodesk and Salesforce. At Autodesk, I implemented new SDK features, reducing non-compliance incidents and technical debt by adhering to best practices in code refactoring. At Salesforce, I optimized SOQL queries and designed custom solutions, improving data processing efficiency and user engagement through robust design and testing. I've also worked on notable projects like UTAP, a Teaching Assistant application portal, and UTMQuest, an online learning tool. These experiences allowed me to enhance my skills in Python, React, PostgreSQL, and other technologies while focusing on high code quality and test coverage. I am passionate about using technology to solve problems and create impactful solutions while following good software engineering principles.

Skills

Proficient: Python, Java, JavaScript, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git, Docker

Competent: React.js, Node.js, MongoDB, Express.js, Pandas

Familiar: Google Cloud Platform, Kubernetes, C/C++, TanStack Query (React Query), Redis

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_ChrisWan

Cluster Monitor [GitHub]: A Minimum viable product to efficiently manage a 10 servers/nodes cluster running Rocky OS. These servers are internally connected through a switch, and are able to communicate through internal IPv4 addresses. Bash scripts were written to facilitate easy monitoring of hardware and usage information for each node in real-time. The collected data is stored in a PostgreSQL (psql) database for reliable data storage. The entire project is deployed within a Docker container to ensure consistent execution and compatibility and development progress is tracked using Git version control. The collected data from this solution will be utilized to generate comprehensive reports, which will inform future resource planning decisions, such as the addition or removal of servers. This will enable the team to optimize resource allocation and ensure efficient cluster management.

Core Java Apps [GitHub]:

- **Grep App:** Grep is a Java-based application offering two efficient approaches for pattern searching within text files. The Stream-Based Approach utilizes Java Streams and functional programming to process large datasets with minimal memory usage, ideal for handling massive files. The Traditional Approach uses conventional file I/O and looping for moderate-sized files. Comprehensive unit testing with JUnit to ensure application correctness. Dockerizing the application facilitates east distribution and deployment.
- **JDBC App:** The JDBC application is a robust system for performing essential CRUD operations (Create, Read, Update, Delete) on a database, utilizing the powerful JDBC API. It integrates a PostgreSQL database within a Docker container environment, with Maven as the package management and dependency resolution tool, and Git as the chosen version control system.

Highlighted Projects

Utap: Contributed to UofT UTAP, a Teaching Assistant application portal, handling 1000+ student applications. Utilized Python Requests to access UofT's timetable API, retrieving and storing 1400+ courses in PostgreSQL. Redesigned flexible test data for a staging environment to meet application workflows achieving 100% test coverage. Engineered custom React modal components to mitigate risks associated with student assignment to conflicting PRA/LEC sessions for professors, reducing assignment errors by 60% and optimizing scheduling processes by 40%.

UTMQuest: Developed an online learning tool with a user-driven question repository, supporting 300+ students. Analyzed 50+ test data and graphs to identify scalable sections. Used Artillery.io for load testing, executed Linux and Bash scripts for system monitoring, and configured lab computers with Virtual Machines for testing. Enhanced database query efficiency by 54% through TanStack Query and Redis integration, while also leveraging MongoDB's skip and limit functionalities for pagination, resulting in a 67% performance enhancement.

NoteIt [GitHub]: Built a web application that summarizes video transcripts using NLP reducing total text by up to 70%, won Best Use of Google Cloud amongst 131 number of participants at a Hackathon (ConuHacks VI).

Professional Experiences

Software Developer Intern, Autodesk (May 2022 - Aug 2022): Implemented and launched new L8 SDK purge features to comply with EU data collection regulations, significantly reducing non-compliance incidents and effectively mitigating potential fines. Led a seamless migration from L8 SDK version 2.0 to 3.0, refactoring over 3,000+ lines of code to enhance maintainability by eliminating deprecated features, resulting in a 30% reduction in technical debt. Debugged and fixed a critical client software issue by implementing permanentRevisionId retrieval upon data revision, resulting in a 50% decrease in client-reported bugs and a 25% improvement in software stability metrics. Developed a User Acceptance Testing (UAT) application leveraging four Autodesk Automation APIs, facilitating testing of over 30 custom and predefined Autodesk JSON models. Increased test coverage by 20% through unit, functional, and integration testing with Jest, Jasmine, and Cucumber.

Software Development Intern, Salesforce (Formerly Traction on Demand) (Sept 2021- May 2022): Optimized Salesforce Queries efficiency by refining SOQL Query and DML statements, utilizing strategic SOQL Indexing and Selective Query Strategies to process over 50,000 records while surpassing heap and DML limits. Designed custom Salesforce solutions, including custom objects, fields, workflows, and validation rules, consistently meeting client-specific requirements and surpassing deadlines. Built a custom front-end component with the Lightning Web Components framework, using JavaScript, React, CSS, and HTML for an engaging UI. Designed and initiated unit testing strategies to uphold code coverage above 80% for projects.

Education

University of Toronto (Sept 2018 - Nov 2023), Bachelor of Computer Science, Mathematical and Computational Sciences

Miscellaneous

- ConuHacks VI Winner - Won Best Use of Google Cloud among 131 participants
- I love breakdancing and was part of GDC (Graydon Dance Club), a breakdancing club in High School
- Played competitive League of Legend in High School
- Love the outdoors and fitness